

ST08

Nanotechnology and New Materials

About 2-3 questions a paper out of the 18-23 that Science and Technology carries. The size definition and the quantum dot/well family are the two near-certain suppliers. Materials - Nitinol, Kevlar, aerogel, Meissner effect - come as a matching question. Treat this as a high-return chapter for the effort.

Nanotechnology is a documented repeat in this exam. RPSC asked it in 2016, 2018, 2021 and 2023 - four papers out of the last five. And it keeps asking the same handful of things: the 1 to 100 nanometre size range, the confinement structures (dot, wire, well), the generations, and who did what. Learn those cold and this chapter pays for itself. The second half - composites, superconductors, semiconductors, rare earths - is newer territory that has grown as India's Semiconductor Mission and critical-mineral push moved into the news.

Nanotechnology and New Materials

Size and scale

- $1 \text{ nm} = 10^{-9} \text{ m} = 10 \text{ angstrom}$
- Nano range = 1 to 100 nm
- At least ONE dimension in range
- Hair is 80,000-100,000 nm
- Greek 'nanos' = dwarf

Why nano is different

- Surface area / volume shoots up
- More atoms sit on the surface
- Reactivity and catalysis rise
- Quantum confinement
- Properties become size-dependent

Confinement family

- Quantum well - confined in 1, free in 2
- Quantum wire - confined in 2, free in 1
- Quantum dot - confined in 3, free in 0
- Dot = 'artificial atom', 2-10 nm
- Nobel Chemistry 2023

Carbon nanomaterials

- Fullerene C60 - 0-D, 1985
- Carbon nanotube - 1-D, 1991
- Graphene - 2-D, 2004
- Diamond, graphite - 3-D bulk
- CNT ~100x stronger than steel

Making and seeing

- Top-down: milling, lithography, etching
- Bottom-up: sol-gel, CVD, self-assembly
- SEM and TEM - electron beam
- AFM and STM - scanning probe
- XRD for crystal structure

Generations (Roco)

- 1st - passive nanostructures
- 2nd - active nanostructures
- 3rd - systems of nanosystems
- 4th - molecular nanosystems

India

- Nano Mission, May 2007, DST
- NSTI 2001 came first
- INST Mohali - dedicated institute
- ARCI Hyderabad, CeNS Bengaluru

New materials

- Composite = matrix + reinforcement
- Superconductor - Meissner effect
- Nitinol - shape-memory alloy
- Metamaterial - negative refraction
- Aerogel - 'frozen smoke'

Strategic materials

- Semicon India - Rs 76,000 crore
- ISM under MeitY
- Dholera - first commercial fab
- Rare earths = 15 lanthanides + Sc + Y
- IREL, monazite beach sands

What 'nano' actually means - start with the number

Begin with arithmetic, because that is what RPSC asks. 'Nano' comes from the Greek nanos, meaning dwarf. One nanometre is one-billionth of a metre - 10^{-9} metre. In the older unit that is 10 angstrom. Nanotechnology is the study and manipulation of matter that has at least one dimension in the 1 to 100 nanometre range. That number, 1 to 100 nm, is the single most asked fact in this chapter. Do not paraphrase it. Learn it as it is.

The scale ladder - fix the size in your head

Object	Approximate size
1 nanometre	10^{-9} metre = 10 angstrom
The nanotechnology range	1 to 100 nm
Width of a DNA double helix	about 2 nm
Quantum dot	about 2 to 10 nm
Typical virus	20 to 400 nm
Red blood cell	about 7,000 nm
Human hair thickness	80,000 to 100,000 nm

- Read the definition carefully - at least ONE dimension must lie in 1-100 nm.
- That is why graphene counts: it is one atom thick even if the sheet is centimetres wide.
- A nanoparticle has all three dimensions in the range; a nanowire has two; a thin film has one.
- Nanotechnology existed before the name - gold nanoparticles gave the Roman Lycurgus Cup its ruby-red glow, and Michael Faraday studied colloidal gold in 1857.

ASKED IN RAS

Asked in RAS Pre 2021 - nanotechnology deals with matter having at least one dimension in which size range? Answer: 1 to 100 nm. The topic also appeared in 2016, 2018 and 2023, which makes it a four-paper repeat.

YAAD RAKHO

Chant it: one nanometre = one-billionth of a metre = 10 angstrom, and the working range is one to a hundred. Three numbers, one line.

Why properties change at the nanoscale - the WHY that now gets tested

Gold is yellow and unreactive in a ring. As a nanoparticle it turns ruby-red and becomes a good catalyst. Same element, different behaviour. Two effects explain this, and RPSC has moved from asking the definition to asking the reason. Learn both by name: the surface-area-to-volume ratio, and quantum confinement.

- Reason 1 - surface area to volume ratio. Break a block into smaller and smaller pieces and the total surface area grows enormously while the amount of matter stays the same.
- So a far larger fraction of the atoms sits on the surface, where bonds are unsatisfied and reactions happen.
- Result: much higher chemical reactivity, stronger catalytic action, better adsorption. This is why nano-catalysts and nano-filters work.
- Reason 2 - quantum confinement. Squeeze a particle down to the size of the electron's wave and the allowed energy levels stop being continuous and become discrete.
- Result: optical and electronic properties become size-dependent. Change only the size and you change the colour of the light emitted.

The confinement family - count the dimensions both ways

Structure	Carriers confined in	Carriers free to move in
Bulk crystal	no direction	3 dimensions (3-D)
Quantum well	1 direction	2 dimensions (2-D)
Quantum wire / nanowire	2 directions	1 dimension (1-D)
Quantum dot	all 3 directions	0 dimensions (0-D)

- A quantum dot is a semiconductor nanocrystal of about 2 to 10 nm, so tightly confined that it behaves like an artificial atom.
- Its emission colour is tunable purely by size - bigger dot, redder light; smaller dot, bluer light.
- Nobel Prize in Chemistry 2023 went to Moungi Bawendi, Louis Brus and Alexei Ekimov for discovering and synthesising quantum dots.
- Quantum dots are used in QLED displays and in biological imaging as fluorescent tags.

ASKED IN RAS

Asked in RAS Pre 2018 - a nanostructure with carriers confined in all three spatial dimensions is called? Answer: quantum dot. RAS Pre 2021 returned to the same family twice: once as a statement question on quantum wells and once as a matching question on free dimensions.

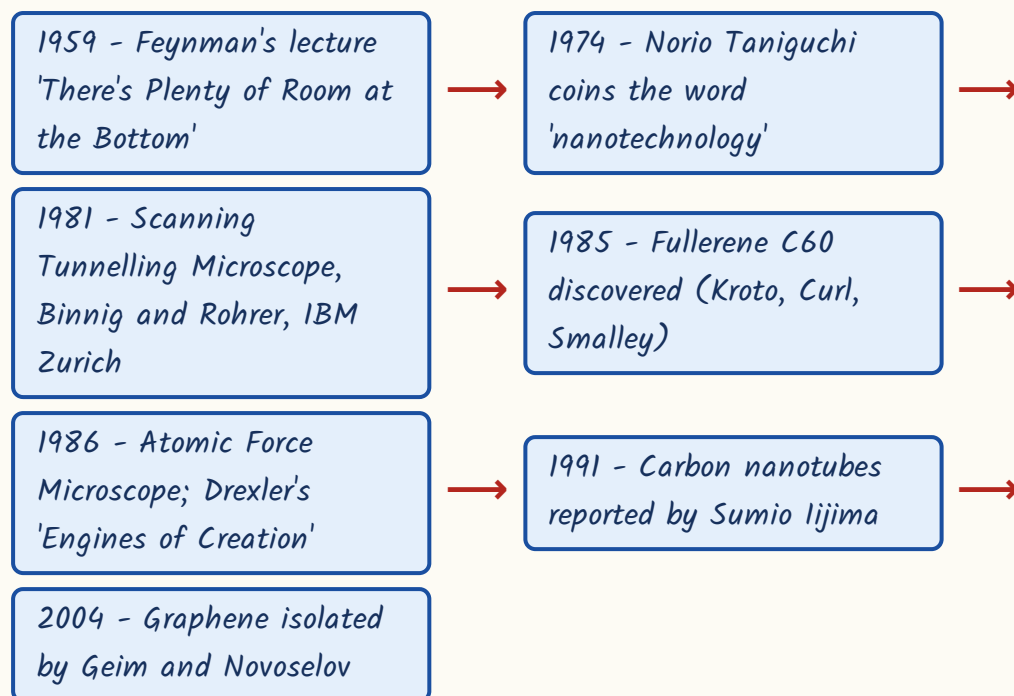
TRAP

Do not swap well and dot. Well = confined in ONE direction. Dot = confined in ALL THREE. The free-dimension count is the mirror image: well 2, wire 1, dot 0. And the colour of a quantum dot DOES depend on its size - a statement saying it does not is the standard wrong option.

The milestones - who did what, in order

This chapter has a clean chronology and RPSC likes ordering questions. Learn the sequence as a story: a physicist imagines it, a Japanese engineer names it, then instruments arrive that can actually see atoms, and then the three carbon forms are discovered one after another - ball, tube, sheet.

The nanotechnology timeline



Nobel Prizes tied to this chapter

Discovery	Who	Prize
Scanning Tunnelling Microscope	Gerd Binnig and Heinrich Rohrer	Physics 1986
Fullerene C60	Harold Kroto, Robert Curl, Richard Smalley	Chemistry 1996
Graphene	Andre Geim and Konstantin Novoselov	Physics 2010
Quantum dots	Bawendi, Brus and Ekimov	Chemistry 2023

- Feynman delivered his lecture on 29 December 1959 at Caltech, to the American Physical Society - the conceptual origin of the field.

- Feynman did not use the word. Taniguchi (Japan) coined 'nanotechnology' in 1974; K. Eric Drexler popularised it in 'Engines of Creation' (1986).
- Graphene was isolated by the adhesive-tape method - peeling layers off graphite until one atomic layer remained.
- Chronological order to remember: Feynman lecture, then C60, then carbon nanotubes, then graphene.

ASKED IN RAS

Asked in RAS Pre 2016 - buckminsterfullerene C60 is made of how many hexagons and pentagons? Answer: 20 hexagons and 12 pentagons. RAS Pre 2023 asked two from this list - who reported carbon nanotubes in 1991 (Iijima) and what the 2010 Physics Nobel was for (graphene).

The nanomaterials - learn them as one table

RPSC asks this as matching: name on the left, description or discoverer on the right. So build the table once and revise it whole. Notice that carbon alone supplies three of the entries, and that the three differ only in how many dimensions they extend in.

Types of nanomaterials

Nanomaterial	What it is	Key point
Nanoparticle	Particle with all dimensions in the nano range	Gold, silver, TiO ₂ , ZnO; carbon black in tyres is a legacy example
Carbon nanotube (CNT)	A rolled-up graphene cylinder	Reported by Iijima, 1991; single-walled and multi-walled
Nanowire	A wire of nanoscale diameter	1-D structure; carriers free in one dimension
Quantum dot	Semiconductor nanocrystal, 2-10 nm	0-D; size-tunable emission; Nobel Chemistry 2023

Nanomaterial	What it is	Key point
Quantum well	Thin layer confining carriers in one direction	2-D; used in laser diodes and LEDs
Fullerene C60	Closed cage of 60 carbon atoms	20 hexagons + 12 pentagons; the 'bucky ball'
Graphene	One-atom-thick honeycomb sheet of sp ² carbon	Isolated 2004; adhesive-tape method
Dendrimer	Repetitively branched, tree-like macromolecule	Drug-delivery carrier; liposomes and nanoshells do the same job

- Carbon by dimensionality: fullerene 0-D, carbon nanotube 1-D, graphene 2-D, diamond and graphite 3-D bulk. Ball - tube - sheet - solid.
- CNTs are about 100 times stronger than steel at roughly one-sixth the weight.
- CNTs are excellent conductors of both heat and electricity.
- CNT electrical behaviour depends on how the sheet is rolled: armchair nanotubes are metallic, zigzag and chiral ones are usually semiconducting.
- So a statement saying 'carbon nanotubes are always insulators' is false - that is a standard wrong option.

TRAP

Nitinol is NOT an allotrope of carbon. It is a nickel-titanium shape-memory alloy. RPSC drops it into a list of diamond, graphite, fullerene and graphene as the odd one out.

Making them and seeing them - synthesis and characterisation

There are only two ways to reach the nanoscale. Either you take a big piece and cut it down, or you take atoms and build up. That is the whole of top-down versus bottom-up. Learn which technique belongs to which side, because the

exam tests exactly that - it puts one technique on the wrong side and asks which statement is correct.

Top-down versus bottom-up synthesis

Top-down	Bottom-up
Start with bulk material and cut it down	Start with atoms and molecules and build up
Sculpting - you remove material	Assembling - you add material
Ball milling, lithography, etching, laser ablation	Sol-gel, chemical vapour deposition, self-assembly, molecular beam epitaxy
Simpler and industrially familiar, but leaves surface defects	Finer control, purer and more uniform structures

TRAP

Chemical vapour deposition is BOTTOM-UP, not top-down. Ball milling is top-down, sol-gel is bottom-up. RPSC's statement question works by flipping exactly one of these three.

Characterisation tools - instrument to principle

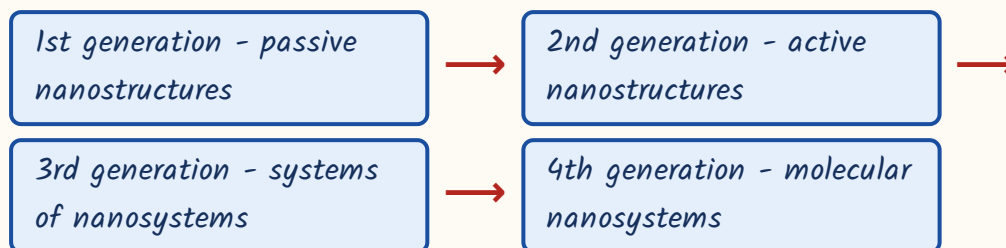
Tool	Working principle	What it gives you
SEM - Scanning Electron Microscope	Scans a focused electron beam across the surface	Surface topography and morphology
TEM - Transmission Electron Microscope	Transmits an electron beam through a very thin specimen	Internal structure and particle size
AFM - Atomic Force Microscope	Measures the force between a sharp tip and the sample surface	3-D surface map; works on non-conducting samples too
STM - Scanning Tunnelling Microscope	Measures the tunnelling current between tip and surface	Atomic-resolution image; sample must conduct
XRD - X-Ray Diffraction	Diffraction of X-rays by the crystal lattice	Crystal structure and phase
DLS - Dynamic Light Scattering	Scattering of light by particles in a liquid	Particle size distribution in suspension

- SEM and TEM are electron microscopes; AFM and STM are scanning probe microscopes. That pairing is the usual matching question.
- The STM (1981) came first and won the 1986 Physics Nobel for Binnig and Rohrer; the AFM followed in 1986.
- Memory hook: the T in STM is tunnelling current, so the sample must conduct. AFM uses Force, so it works on plastics and biological samples as well.

The four generations of nanotechnology

This framework is credited to M.C. Roco and it is pure matching material - RPSC asked it as a four-pair match in 2018. The logic is easy: the technology starts with materials that just sit there, then moves to materials that do something, then to whole systems, and finally to building molecule by molecule.

Generations of nanotechnology (Roco)



- 1st - passive: fixed function. Coatings, nanoparticles in sunscreen and paint, reinforced polymers.
- 2nd - active: the structure changes its behaviour while in use. Targeted drug carriers, adaptive structures, actuators.
- 3rd - systems of nanosystems: many nanocomponents working together, guided assembly, three-dimensional networking.
- 4th - molecular nanosystems: design and assembly molecule by molecule; still largely a research goal.

ASKED IN RAS

Asked in RAS Pre 2018 as a match - first generation to passive nanostructures, second to active nanostructures, third to systems of nanosystems, fourth to molecular nanosystems. Learn the order, not the explanations.

YAAD RAKHO

P-A-S-M: Passive, Active, Systems, Molecular. Four generations, four letters, in order.

Applications, nanotoxicology and India's Nano Mission

Applications come as 'which of these is NOT an application of nanotechnology'. So you do not need depth here - you need coverage, sector by sector, so that nothing on the list looks unfamiliar. Note the one clean negative: generating electricity by nuclear fission has nothing to do with nanotechnology.

Nanotechnology by sector

Sector	Nanotechnology at work
Medicine	Targeted drug delivery using dendrimers, liposomes and nanoshells; nano-scale imaging and diagnostics
Cosmetics	Nano-TiO ₂ and nano-ZnO as transparent UV blockers in sunscreen
Textiles	Stain-resistant and water-repellent finishes; nano-silver as an antimicrobial
Water	Nano-membranes and nano-filters for purification and desalination
Energy	Higher-capacity battery electrodes, better solar cell materials, fuel cell catalysts
Electronics	Ever smaller transistors, quantum-dot displays, flexible electronics
Packaging	Nano-clay barrier films that keep out oxygen and moisture
Agriculture	Nano-fertilisers and controlled-release pesticides
Defence	Lighter and stronger composite armour, sensors, protective coatings

- Nanotoxicology is the study of the harm nanomaterials can do - and it is a fair exam target because policy documents stress it.
- Inhaled ultrafine particles are small enough to cross biological barriers that stop ordinary dust.
- Long, thin carbon nanotubes have raised asbestos-like concerns in animal studies.
- Because of this, risk assessment, labelling and regulation are built into national nanotechnology programmes.

India's programme - dates and the parent department

- The Nano Science and Technology Initiative (NSTI) came first, in 2001.
- The Nano Mission was launched in May 2007 under the Department of Science and Technology - remember DST, not MeitY.
- It concluded on 31 March 2017 and continues as the National Programme on Nano Science and Technology.
- Institute of Nano Science and Technology (INST), Mohali is DST's dedicated autonomous nano institute.
- Other key centres: ARCI, Hyderabad (powder metallurgy and new materials) and CeNS, Bengaluru (nano and soft matter sciences).

New materials - composites, superconductors and smart materials

The second half of this chapter is a vocabulary test. Each material has one defining property and one signature use, and the question is always a match. Do not read around it - just fix the pairs. The one entry worth a second reading is the metamaterial, because its behaviour comes from the engineered structure and not from the chemistry - that wording itself has been made into a statement question.

- A composite = matrix + reinforcement. Two materials combined so that the result behaves better than either alone.
- Carbon-fibre reinforced polymer (CFRP) - carbon fibres in a polymer matrix; used in aircraft and sports goods.
- Kevlar is an aramid fibre - the bullet-proof-vest material.
- Fibreglass is glass fibre in a resin matrix. Reinforced concrete is steel in concrete - the oldest everyday composite.
- The point to remember: a composite's properties differ from those of either constituent taken separately.

Superconductors

- Superconductivity = zero electrical resistance below a critical temperature, PLUS expulsion of magnetic flux from inside the material.
- That expulsion is the Meissner effect - the name RPSC asks for.
- Discovered by Heike Kamerlingh Onnes in mercury in 1911.
- Everyday use: the powerful magnets in MRI scanners; research use: maglev transport and lossless power transmission.

Smart materials, metamaterials and aerogel

Material to property to use - the standard matching set

Material	Type	Signature use
Nitinol (nickel-titanium)	Shape-memory alloy - returns to a remembered shape on heating	Vascular stents, spectacle frames
Quartz, PZT	Piezoelectric - produce a voltage under mechanical stress and vice versa	Sensors, quartz watches, gas lighters
Thermochromic materials	Change colour with temperature	Temperature indicators, novelty coatings
Magnetostrictive materials	Change shape in a magnetic field	Actuators and sonar transducers

Material	Type	Signature use
Aerogel	Extremely low-density solid, popularly 'frozen smoke'	Among the lightest solids; thermal insulation
Metamaterial	Behaviour comes from engineered structure, not chemistry	Negative refractive index; cloaking and superlens research

Semiconductors and rare earths - the strategic materials

This is the newest part of the chapter and the part most likely to be asked as current affairs. Two things to hold: the physics of doping, which is old NCERT material, and India's semiconductor and critical-mineral programmes, which are current. Both are short and both are examinable.

- A semiconductor's conductivity sits between a conductor's and an insulator's; silicon's band gap is about 1.1 eV.
- Doping with a group-15 element such as phosphorus or arsenic adds free electrons - that gives an n-type semiconductor.
- Doping with a group-13 element such as boron or gallium creates holes - that gives a p-type semiconductor.
- Trick: n for negative carriers comes from group 15 (five valence electrons); p for positive holes comes from group 13 (three).
- Silicon carbide (SiC) and gallium nitride (GaN) are wide-band-gap semiconductors, used in high-power and high-temperature electronics.

India's semiconductor push - the facts that get asked

Item	Detail
Programme	Semicon India Programme, approved 2021, outlay Rs 76,000 crore
Who runs it	India Semiconductor Mission (ISM), under the Ministry of Electronics and Information Technology (MeitY)

Item	Detail
First commercial fab	Tata Electronics with Powerchip (PSMC) at Dholera, Gujarat
Assembly and testing units	Micron's ATMP and Kaynes' unit at Sanand, Gujarat; Tata's OSAT plant at Jagiroad, Assam

CURRENT LINK

Current link, as of September 2026: per PIB (1 April 2026) ten projects worth about Rs 1.6 lakh crore stood approved under the India Semiconductor Mission. ISM 2.0 was announced in Budget 2026-27 and cleared by the Union Cabinet at the end of June 2026 with an outlay of roughly Rs 1.25 lakh crore. For the exam, the two safest facts remain the ministry (MeitY) and the first fab location (Dholera).

- Rare earths are 17 elements - the 15 lanthanides plus scandium and yttrium.
- They are not actually rare in the crust; they are rarely found in concentrated, economically workable deposits.
- India's principal source is monazite in beach sands - Kerala, Odisha, Tamil Nadu and Andhra Pradesh.
- IREL (India) Limited is the principal public sector producer of rare earths in India.
- The National Critical Mineral Mission was launched in January 2025 to secure supply of these and other critical minerals.

REVISION RECAP

1. 1 nanometre = 10^{-9} metre = 10 angstrom; nanotechnology = at least one dimension in 1-100 nm. Asked 2016, 2018, 2021, 2023.
2. Two nanoscale effects: surface-area-to-volume ratio (higher reactivity) and quantum confinement (size-dependent optical and electronic behaviour).
3. Confined in 1 = quantum well (free in 2); confined in 2 = quantum wire (free in 1); confined in 3 = quantum dot (free in 0, an artificial atom).

4. Quantum dot: semiconductor nanocrystal 2-10 nm, colour tunable by size, Nobel Chemistry 2023 - Bawendi, Brus, Ekimov.
5. Feynman 1959 lecture, Taniguchi coined the word 1974, Drexler popularised it 1986.
6. Fullerene C₆₀ = 20 hexagons + 12 pentagons, 1985, Kroto-Curl-Smalley, Chemistry Nobel 1996.
7. Carbon nanotubes - Iijima 1991; about 100 times stronger than steel at one-sixth the weight; armchair metallic, zigzag/chiral semiconducting.
8. Graphene - one-atom-thick honeycomb sheet, Geim and Novoselov 2004, Physics Nobel 2010.
9. Carbon by dimension: fullerene 0-D, CNT 1-D, graphene 2-D, diamond and graphite 3-D.
10. Top-down = ball milling, lithography, etching, laser ablation. Bottom-up = sol-gel, CVD, self-assembly, MBE.
11. SEM and TEM are electron microscopes; AFM and STM are scanning probe. STM built 1981 by Binnig and Rohrer, Physics Nobel 1986.
12. Generations: 1st passive, 2nd active, 3rd systems of nanosystems, 4th molecular nanosystems - Roco. Matched in RAS 2018.
13. Dendrimers, liposomes and nanoshells are drug-delivery carriers; nano-silver is antimicrobial; nano-TiO₂ and ZnO block UV.
14. Nanotoxicology: ultrafine particles cross biological barriers; long thin CNTs raise asbestos-like concerns.
15. Nano Mission - May 2007, under DST; NSTI 2001 preceded it; concluded 31 March 2017 and continues as the National Programme. INST Mohali is the dedicated institute.
16. Composite = matrix + reinforcement: CFRP, Kevlar (aramid), fibreglass, reinforced concrete.
17. Superconductivity = zero resistance below the critical temperature + Meissner effect; Kamerlingh Onnes, mercury, 1911.
18. Nitinol = shape-memory alloy (stents); quartz and PZT = piezoelectric; aerogel = 'frozen smoke' insulator; metamaterial = negative refractive index.
19. Semicon India 2021, Rs 76,000 crore; India Semiconductor Mission under MeitY; first commercial fab at Dholera by Tata with PSMC.
20. Rare earths = 15 lanthanides + scandium + yttrium; monazite beach sands; IREL is the PSU producer; National Critical Mineral Mission, January 2025.